

MAX TRAN

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P.C. Rossin College of Engineering & Applied Science
Lehigh University

May 2026
Bethlehem, PA

Major: Bachelor of Science in Biocomputational Engineering

Coursework: Fundamentals of Biomedical Signals; BIOE Applications in ML; Biomedical Image Computing

WORK EXPERIENCE

Summer Intern

May 2025 - August 2025

X9, Inc

Wallingford, CT

- Designed and trained deep learning models in PyTorch including CNN14 and CNN6 to model physiological signal representations for auscultation-based stenosis detection, achieving 93% sensitivity and 86% specificity
- Built an end-to-end machine learning pipeline, including signal preprocessing, feature extraction, model training, validation, and performance evaluation
- Implemented a SimCLR-style contrastive pretraining framework to learn latent representations from limited labeled biomedical signals, enabling performance in low-data regimes
- Designed and executed controlled experimental comparisons across architectures and training strategies to evaluate robustness, generalization, and feasibility for translational clinical use

Operations Intern

May 2023 - August 2023

Revelle Aesthetics, Exploramed

Mountain View, CA

- Analyzed manufacturing and testing data using Minitab to identify sources of error and improve process reliability
 - Developed a Python based program utilizing QR codes to transfer inventory data into Excel for automated tracking and reporting
 - Gained familiarity with ISO 13485 standards and regulatory considerations in medical device development
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PROJECTS

Gastric Cancer RNA-seq Differential Expression Analysis

Python, PyDESeq2, scikit-learn, gseapy

- Built an end-to-end RNA-seq analysis pipeline on TCGA gastric cancer data (448 samples), identifying 11,727 differentially expressed genes between tumor and normal tissue using DESeq2
- Trained a Random Forest classifier on top differentially expressed genes, identifying and correcting a data leakage issue in feature selection to achieve a rigorously validated ROC-AUC of 1.0 on held-out samples

Machine Learning Algorithm for Detecting Skin Cancer Using VOCT Imaging - Optovibronex

Google Colab, Python, PyTorch

- Prototyped and trained a CNN-based models to interpret biomedical imaging data (Vibrational Optical Coherence Tomography) for noninvasive skin cancer detection, achieving 90% sensitivity and specificity through hyperparameter tuning and transfer learning
 - Preprocessed and augmented imaging data to improve model generalization and performance
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SKILLS:

PyTorch, scikit-learn, Convolutional Neural Networks (CNNs), Self-Supervised Learning (SimCLR), Python, Java, SQL, NumPy, pandas, Matplotlib, Git/GitHub, statistical evaluation, model comparison, performance metrics, PyDESeq2, bioinformatics, differential expression analysis